

Afreen Gul

Machine Learning Engineer | Computer Vision | Data Science

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PROFESSIONAL SUMMARY

Motivated Machine Learning Engineer and Computer Science undergraduate (CGPA 3.95/4.00) with proven hands-on experience in end-to-end model development, computer vision, and data-driven problem solving. Proficient in Python, TensorFlow, Scikit-learn, and OpenCV. Skilled in designing, training, evaluating, and optimising machine learning and deep learning models on real-world datasets. Committed to delivering scalable, accurate, and ethical AI solutions that create measurable business and research impact.

TECHNICAL SKILLS

Programming Languages: Python, HTML, CSS, JavaScript

Machine Learning: Supervised Learning, Unsupervised Learning, Regression, Classification, Anomaly Detection, Model Training & Evaluation

Deep Learning & CV: Convolutional Neural Networks (CNN), YOLO, Object Detection, Image Classification, Real-Time Inference

Libraries & Frameworks: TensorFlow, Keras, Scikit-learn, OpenCV, NumPy, Pandas, Matplotlib, Seaborn

Data Science: Exploratory Data Analysis (EDA), Feature Engineering, Data Preprocessing, Data Visualisation, Statistical Analysis

Tools & Platforms: Git, GitHub, Jupyter Notebook, Google Colab, VS Code, Git & Github

Soft Skills: Analytical Thinking, Problem Solving, Research, Teamwork, Communication, Continuous Learning

PROJECTS

Sign Language Digit Recognition | Python · TensorFlow · OpenCV · CNN

- Trained a CNN to classify hand-sign digits from images with real-time webcam inference via OpenCV.
- Applied data augmentation and tuning to handle varied lighting and hand orientations.

Credit Card Fraud Detection | Python · Scikit-learn · Pandas · SMOTE

- Built a classification model on an imbalanced dataset; applied SMOTE to improve minority-class recall.
- Evaluated with F1-score and ROC-AUC, reducing false negatives to minimise financial risk.

House Price Prediction | Python · Scikit-learn · Pandas · Matplotlib

- Developed and compared Regression models (Linear, Ridge, Lasso) on Kaggle housing data.
- Feature engineering and cross-validation reduced RMSE; results visualised for clear reporting.

EDUCATION

Bachelor of Science in Computer Science

University of Engineering and Technology (UET), Taxila 2023 – 2027 | CGPA: 3.95 / 4.00

Relevant Coursework: Machine Learning, Artificial Intelligence, Data Structures & Algorithms, Computer Vision, Database Systems, Software Engineering.

Intermediate in Computer Science (ICS)

FG Post Graduate College For Women, Wahcantt 2021 – 2023 | Grade: A+

Secondary School Certificate (SSC / Matriculation)

FG Public School HIT Taxila Cantt 2019 – 2021 | Grade: A+

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

Machine Learning with Python — *IBM / Coursera*

Python for Data Science, AI & Development — *IBM / Coursera*

Project Development in Python — *Microsoft / Coursera*

Introduction to Artificial Intelligence — *Google / Coursera*

The Art of Prompting — *Google / Coursera*

Responsible AI Practices — *Google / Coursera*

KEY COMPETENCIES

Machine Learning • Deep Learning • Computer Vision • Artificial Intelligence • Predictive Modeling • Neural Networks • Transfer Learning • Data Analysis • Feature Engineering • Model Deployment • Classification • Regression • Clustering • Anomaly Detection • Statistical Analysis • Data Preprocessing • Python Programming • Research & Development • Algorithm Optimization • Natural Language Processing • Agile Methodology

LANGUAGES

English (Professional Working Proficiency) **Urdu** (Native / Full Professional Proficiency)